

CICT1105: Computer and Communication Technology

ICT Self-Study Guide

Topic 1:

1. Basic Definitions in ICT

Goal: Understand key ICT terms clearly and accurately.

Step 1: Start with Simple Definitions

Write clear definitions of the following terms:

- **Computer**
- **Information**
- **Communication**
- **Technology**
- **Internet**
- **Cloud Computing**
- **Data**
- **Communication Technology**
- **Information Technology**
- **Information and Communication Technology (ICT)**

Test Yourself

- a. How would your daily academic life change if you had to perform all tasks without a computer? Give three specific examples.
- b. Computers are described as “fast and accurate.” Describe a scenario where a computer might produce incorrect results and explain why.
- c. How does raw data become useful information? Give a real example.
- d. Can the same data lead to different information depending on who reads it? Why?
- e. What makes information trustworthy or valuable

- f. How does the choice of communication channel (text, voice, video) change the message's meaning?
- g. What barriers can stop effective communication in a classroom or online? How would you solve them?
- h. How can poor communication lead to wrong decisions? Give an example.
- i. How can the same technology be used for good and for harm? Give examples.
- j. In what ways has a specific technology (e.g., smartphones) changed learning at your school?
- k. What makes a technology sustainable and appropriate for a community?
- l. How does ICT change the way people learn, work, and interact in your community?
- m. What are the main risks (privacy, misuse, access) of ICT, and how can they be reduced?
- n. How could ICT be used to solve one major problem at your school (e.g., attendance tracking)?
- o. **Imagine you are working remotely in a team spread across different countries.**
What Internet tools and cloud applications would improve collaboration and productivity, and why?
- p. **Internet speeds vary greatly between rural and urban communities.**
How does this inequality affect economic development and access to digital services?
- q. **Cloud computing uses data centres that run 24/7.**
Discuss the environmental impact of cloud computing and suggest ways the ICT industry can reduce energy consumption.

2. The Role of Computers in ICT

Goal: Understand how computers are used in communication and information processing.

Step 1: Explore Common Computer Uses

Make a list of where and how computers are used in daily life:

- **At School:** Typing assignments, internet research etc.
- **At Home:** Watching videos, social media etc.
- **At Workplaces:** Managing data, designing, emailing etc.

Tip: with your classmate discuss how your own phone or laptop is used every day.

Step 2: Use a Computer for a Real Task Practice

these activities:

- Open a Word Processor (e.g., MS Word or Google Docs)

Tip: This shows how a computer helps in creating and storing information.

Step 3: Try Communication Tools

Use a computer or smartphone to:

- Send an email
- Chat on WhatsApp or Messenger
- Join a Zoom or Google Meet session

Tip: Take note of what device and software you used to communicate.

Step 4: Make a Flowchart or Diagram

Draw a simple diagram that shows how computers help with:

- **Collecting data**
- **Processing data**
- **Storing data**
- **Communicating data**

Tip: Use arrows and icons to make it visual and easy to understand.

Step 5: Reflect

Answer these reflection questions:

- a. How would communication in modern workplaces change if computers were removed from all ICT systems for one week? Provide specific examples.

- b.** Explain how computers improve decision-making in organisations. Provide a real-world example of a poor decision that could result from a lack of computerised data analysis.
- c.** In what ways have computers transformed teaching and learning processes in higher education? Evaluate both positive and negative impacts.
- d.** Analyse how computers support effective management of large institutions such as Mzuzu University. Which ICT functions would collapse without them?
- e.** Do computers always improve efficiency? Describe a scenario where using a computer might slow down work instead of speeding it up.
- f.** How do computers help reduce human error in ICT operations? Give two examples where errors still occur even with computer-based systems.
- g.** Assess the impact of computers on communication speed, accuracy, and cost. Which of the three is most improved and why?
- h.** Why are computers essential in network management and troubleshooting? Provide examples of tools and tasks that depend entirely on computers.
- i.** Evaluate the importance of computers in information storage. What would be the consequences of storing all university records on paper instead of in digital form?
- j.** How do computers enable automation in ICT? Identify two jobs that became easier and two that became obsolete due to automation.
- k.** If computers can process large amounts of data quickly, why do some organisations still struggle with poor data management? Explain with examples.
- l.** Discuss how computer viruses and cyber threats affect the role of computers in ICT. Can computers still be considered reliable despite these risks?
- m.** How do mobile computers (e.g., smartphones, tablets) change the way ICT services are delivered in remote areas?
- n.** Imagine you are designing an ICT system for a hospital. Why would computers be essential, and what risks come with relying heavily on them?

- o.** In what ways do computers help integrate various ICT systems (e.g., LMS, SMS, communication tools) in universities?
- p.** Compare the role of computers in ICT today with their role 20 years ago.
What major transformations have taken place?
- q.** How do computers support real-time communication in organisations? Give an example where real-time communication prevented a major problem.
- r.** Explain how computers facilitate information sharing among departments.
Why might some departments still resist ICT-based workflows?
- s.** What is the relationship between computer speed and the quality of ICT services delivered? Can slow computers affect customer satisfaction? How?
- t.** Predict the role of computers in ICT 20 years from now. How might emerging technologies like AI, IoT, and cloud computing reshape ICT roles?